

PROJECT DETAILS · FEATURES · DEMO SCREENS

Agentic AI Smart City Ecosystem

Two AI systems for the Municipal Corporation, citizens and police,
joined by a privacy-preserving data bridge

Team

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Guide

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Version 2 · October 2026 · adds mobile app, system connectivity and privacy-by-design police video

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2026–27

1. Project Overview

1.1 Problem

- Municipal departments (water, roads, waste, electricity) and the police work separately. Coordination happens through calls and letters.
- Complaints are handled first-come-first-served, not by severity. Many people report the same issue again and again.
- Officers can mark complaints "resolved" without proof, and citizens rarely get updates.
- Citizens get no advance notice of water cuts, power shutdowns or road closures, so they file complaints that could have been avoided.
- Police CCTV sees accident hotspots and traffic congestion every day, but this knowledge never reaches the people who plan roads and junctions.

1.2 Proposed Solution

An **agentic AI platform** made of **two independent systems**:

System 1: Municipal Corporation + Citizens

Complaint reporting, AI routing and prioritization, officer → ward rep → mayor accountability, ward announcements, public statistics and infrastructure insights.

System 2: Police Control Room

AI video analytics on CCTV to detect accidents, congestion, crowds and fights, with live alerts and dispatch. Runs on a separate, secure system.

Privacy-by-design: live view only, and video is stored only with supervisor approval.

Privacy Data Bridge: a one-way link that sends only **aggregated, anonymized** accident and congestion statistics and road events from the police system to the municipal system. **No video, faces or number plates** are ever shared.

1.3 Dashboards

Dashboard	Views / Users	Purpose
1. City Authority (Municipal Corporation)	Dept. Officer · Ward Representative · Mayor	Complaint handling, escalation, announcements, city analytics, infrastructure planning
2. Police Control Room	Police control room operators	Live CCTV monitoring (no recording by default), AI incident alerts, dispatch, evidence approval, audit log
3. Citizen App	Public (Android mobile app)	Photo complaints, status tracking, ward announcements, feedback
Common: Public Statistics	Citizens + authorities	Complaints received vs resolved, by ward and department

2. System Architecture

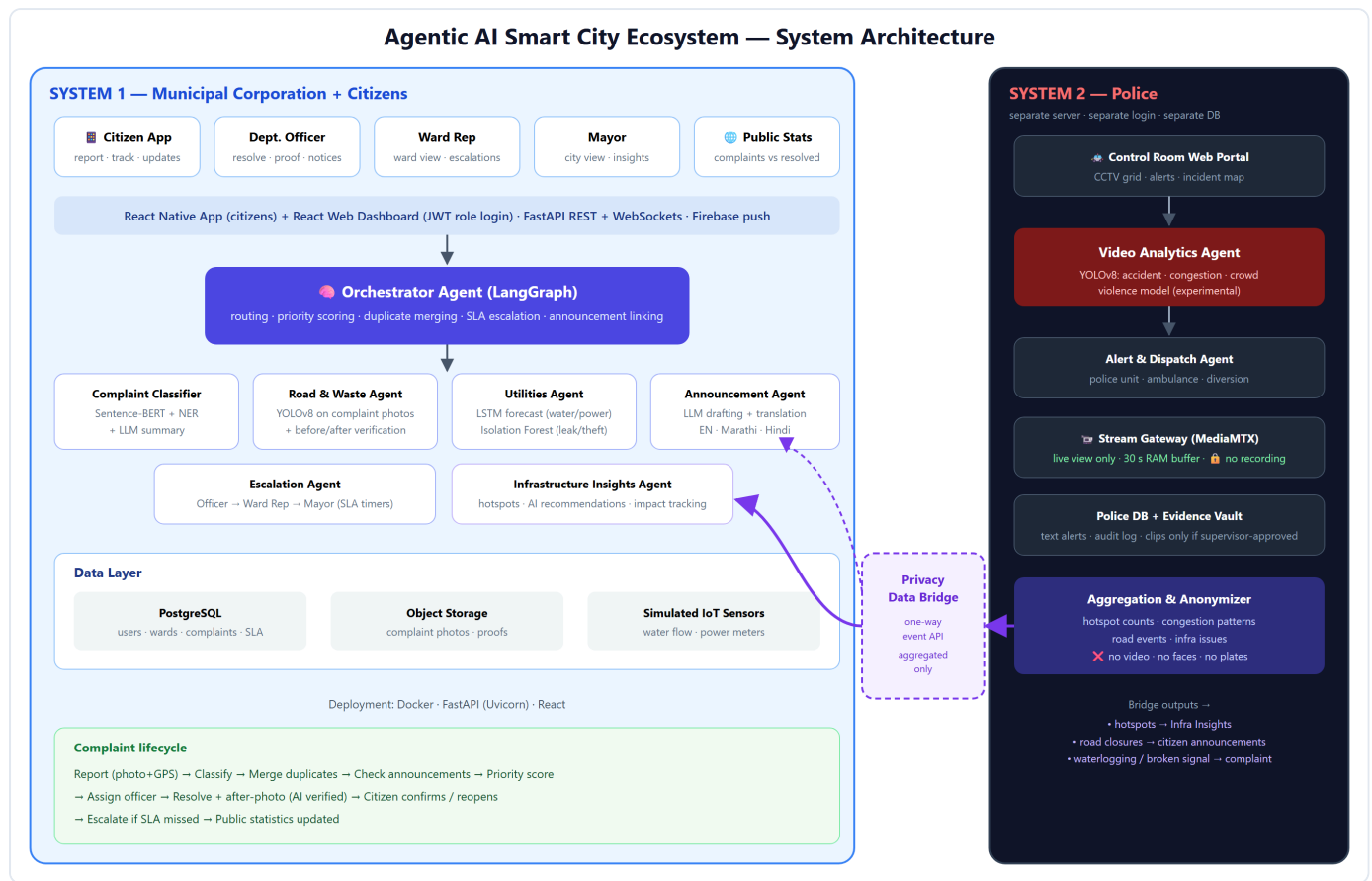


Figure 1: System architecture. Two independent systems connected by a one-way privacy data bridge.

- **Frontend:** a **React Native (Expo) mobile app** for citizens (Android APK), and React web dashboards for the municipal corporation and police.
- **Backend:** FastAPI (Python) REST APIs, with WebSockets for real-time updates.
- **AI layer:** a LangGraph Orchestrator Agent that coordinates specialized agents.
- **Data:** PostgreSQL, object storage for photos, and simulated IoT sensor streams.
- **Police system:** its own server, database and login on the police network. Live video is relayed without recording, and it shares data only through the anonymizer and the one-way bridge.

2.1 How the Systems Connect

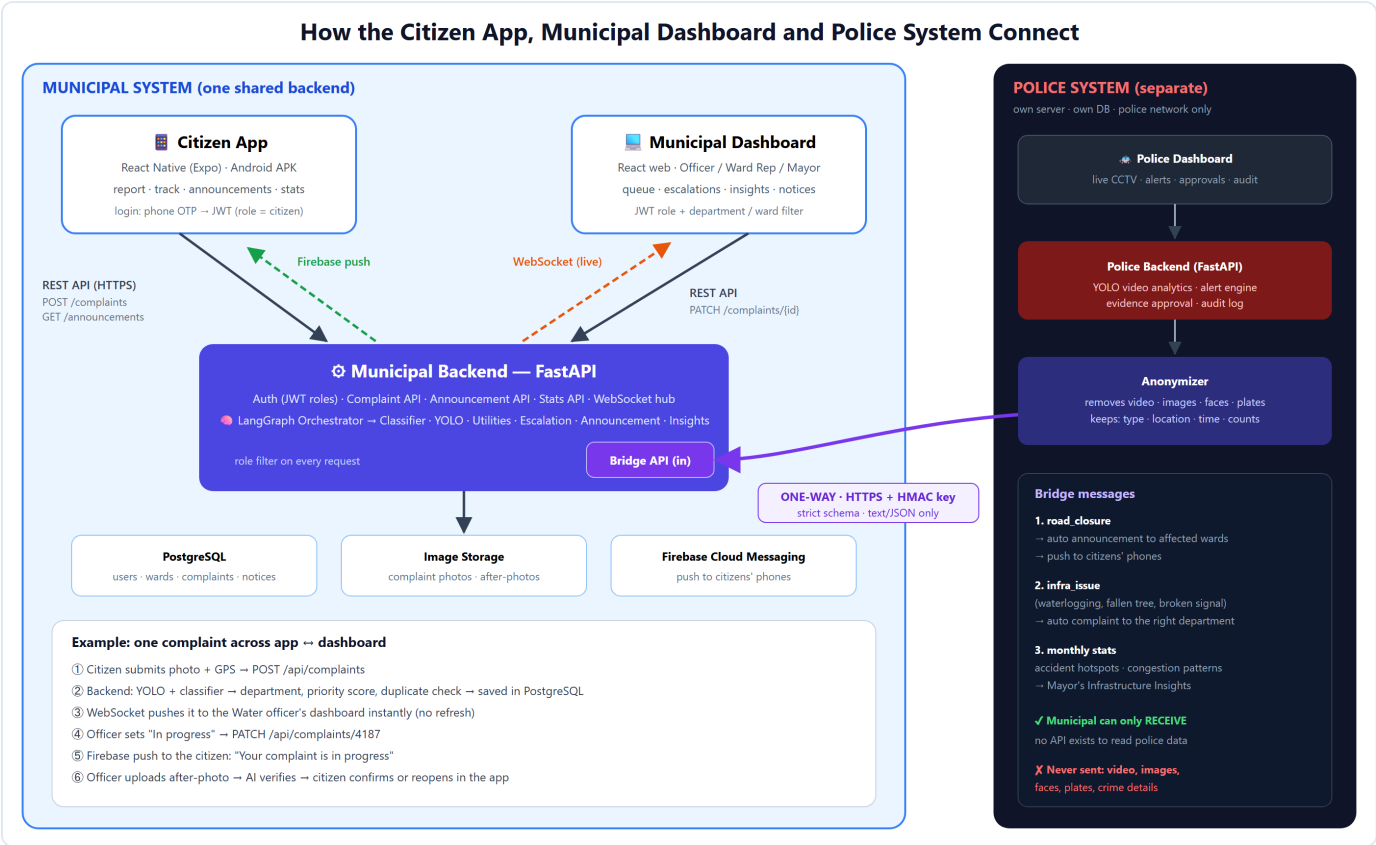


Figure 2: How the citizen app, municipal dashboard and police system connect.

The **citizen app** and the **municipal dashboard** **never talk to each other directly**. Both are clients of the **same municipal backend and database**, so whatever one side changes, the other sees. The **police system** is **separate** and can only **send** anonymized text events to the municipal backend through a one-way Bridge API.

Connection	Technology	Used for
Citizen App → Backend	REST API (HTTPS) + JWT	Submit complaints with photo and GPS, fetch status, announcements and statistics
Backend → Citizen App	Firebase push notifications	Status updates and announcements, even when the app is closed
Dashboard → Backend	REST API (HTTPS) + JWT	Assign, update and resolve complaints, upload proof, post announcements
Backend → Dashboard	WebSocket	New complaints and escalations appear instantly, without refreshing
Police → Municipal	One-way Bridge API (HTTPS + HMAC key)	Road closures, infrastructure issues and monthly statistics, as text/JSON only

2.2 Role-Based Access

At login every user receives a **JWT token** carrying their role, for example `{"role": "officer", "department": "water"}`. The backend applies a filter to every request:

Role	Data visible
Citizen	Only their own complaints, their ward's announcements, public statistics
Dept. Officer	Complaints of their department
Ward Representative	All complaints in their ward
Mayor	Entire city, plus infrastructure insights

2.3 Main APIs

Endpoint	Purpose
POST /api/auth/otp	Citizen login with phone OTP
POST /api/complaints	New complaint (photo + GPS + description)
GET /api/complaints/mine	Citizen's own complaints
POST /api/complaints/{id}/feedback	Rate or reopen
GET /api/complaints?dept=water	Officer's priority queue
PATCH /api/complaints/{id}	Assign or update status
POST /api/complaints/{id}/proof	Upload after-photo for AI verification
GET/POST /api/announcements	Read ward announcements / post new ones
GET /api/stats/public	Public statistics
GET /api/insights/hotspots	Infrastructure insights from bridge data
WS /ws/dashboard	Live dashboard updates
POST /bridge/events, POST /bridge/stats	Receive-only endpoints for police data

2.4 Bridge Message Example

```
POST /bridge/events      (from police server, signed with HMAC key)
{
  "type": "road_closure",
  "location": { "lat": 16.6968, "lng": 74.2433, "area": "Dabholkar Corner" },
  "affected_wards": [11, 12],
  "reason": "accident",
  "until": "2026-10-01T18:00:00"
}
```

→ Announcement Agent publishes "Road closed near Dabholkar Corner till 6 PM" to Wards 11 and 12, and citizens receive a push notification.

Bridge security: HTTPS encryption · HMAC/API-key authentication so only the police server can send · strict Pydantic schema that rejects any image or video field · every message logged · the municipal side has **no API to read police data**.

3. System 1: Municipal Corporation + Citizens

3.1 Citizen App

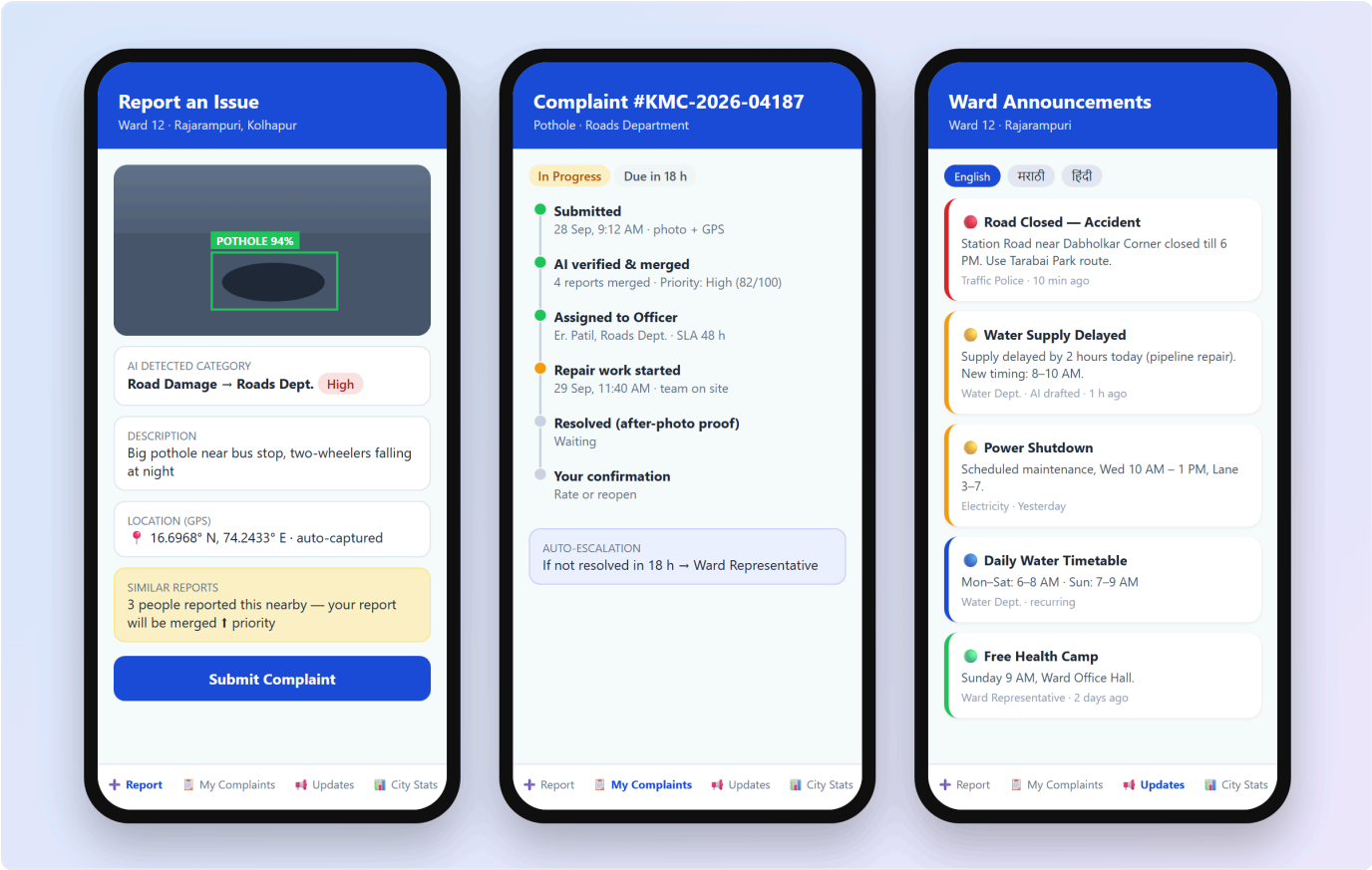


Figure 3: Citizen app. Report an issue, track a complaint, ward announcements.

Platform: native mobile app built with **React Native (Expo)**, shared as an Android APK. It uses the same React skills as the web dashboards. Key libraries: expo-image-picker (camera), expo-location (GPS), expo-notifications + Firebase (push), i18next (EN / Marathi / Hindi).

Feature	Description
Photo complaint	Click a photo and add a short description. GPS location and ward are captured automatically.
AI auto-categorization	YOLOv8 + Sentence-BERT find the issue type (pothole, garbage, leakage...) and the department. No manual category selection.
Duplicate detection	If similar reports exist nearby, the new one is merged and the incident's priority increases.
Complaint tracking	Timeline: Submitted → Verified → Assigned → In progress → Resolved (with after-photo) → Citizen confirmation.
Rate / Reopen	The citizen rates the resolution or reopens the complaint if it isn't actually fixed.
Ward announcements	Water supply timings, power cuts, road closures, events. Available in English, Marathi and Hindi.
Push notifications	Status changes and important announcements (Firebase Cloud Messaging; SMS for emergencies).
City statistics	Access to the public transparency page.
Login	Phone number + OTP; ward detected from the address or GPS.

3.2 Department Officer View

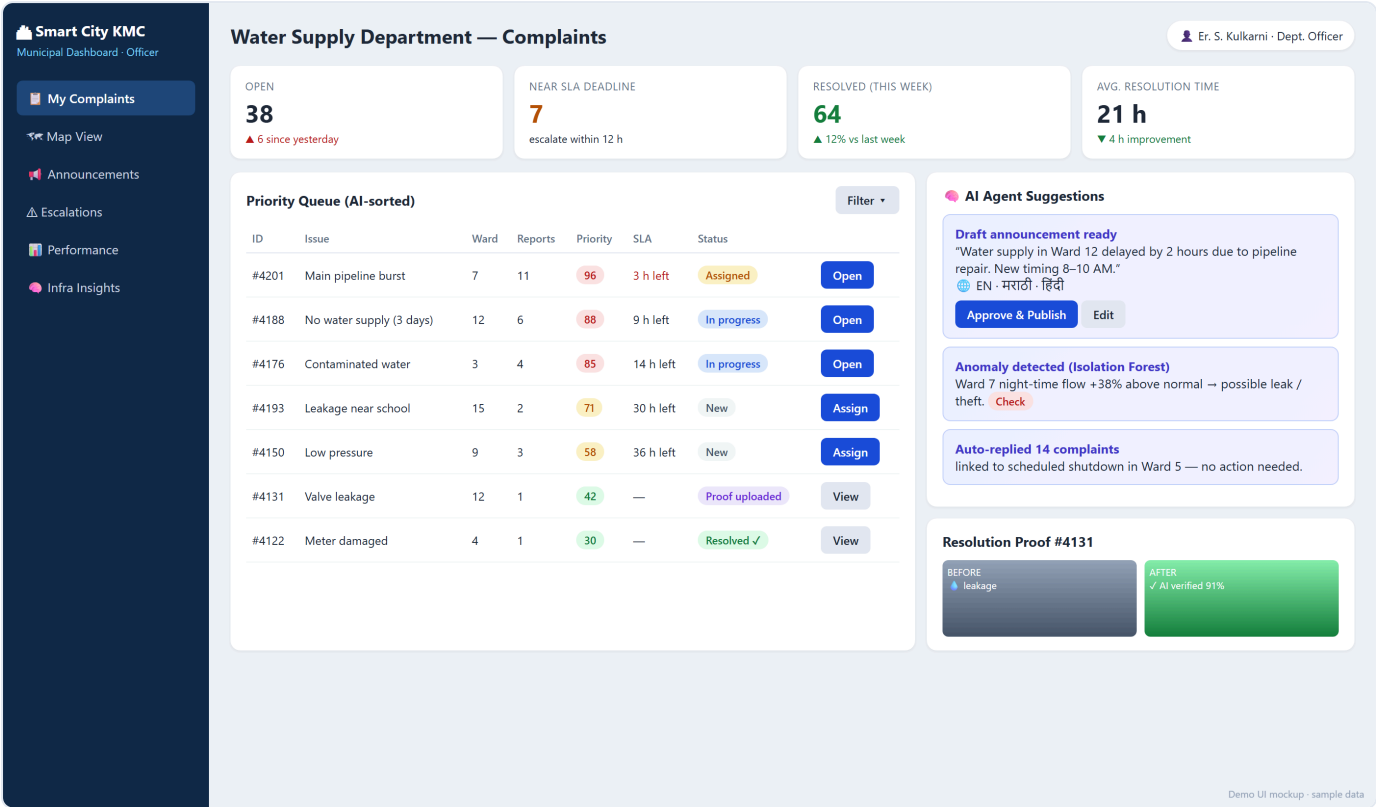


Figure 4: Department Officer view (Water Supply). AI-sorted priority queue and agent suggestions.

- **AI-sorted priority queue:** complaints ranked by a priority score (0–100), not by arrival time.
- **SLA timers:** each complaint has a deadline. Complaints close to it are highlighted.
- **Assign and update:** assign field staff and change the status.
- **Resolution proof:** upload an after-photo. AI compares it with the before-photo before the complaint can be closed.
- **AI suggestions:** announcement drafts ready for one-click approval, anomaly alerts (possible leak or theft), and auto-replies to complaints explained by a scheduled shutdown.
- **Post announcements** for the department's service areas.

3.3 Ward Representative View

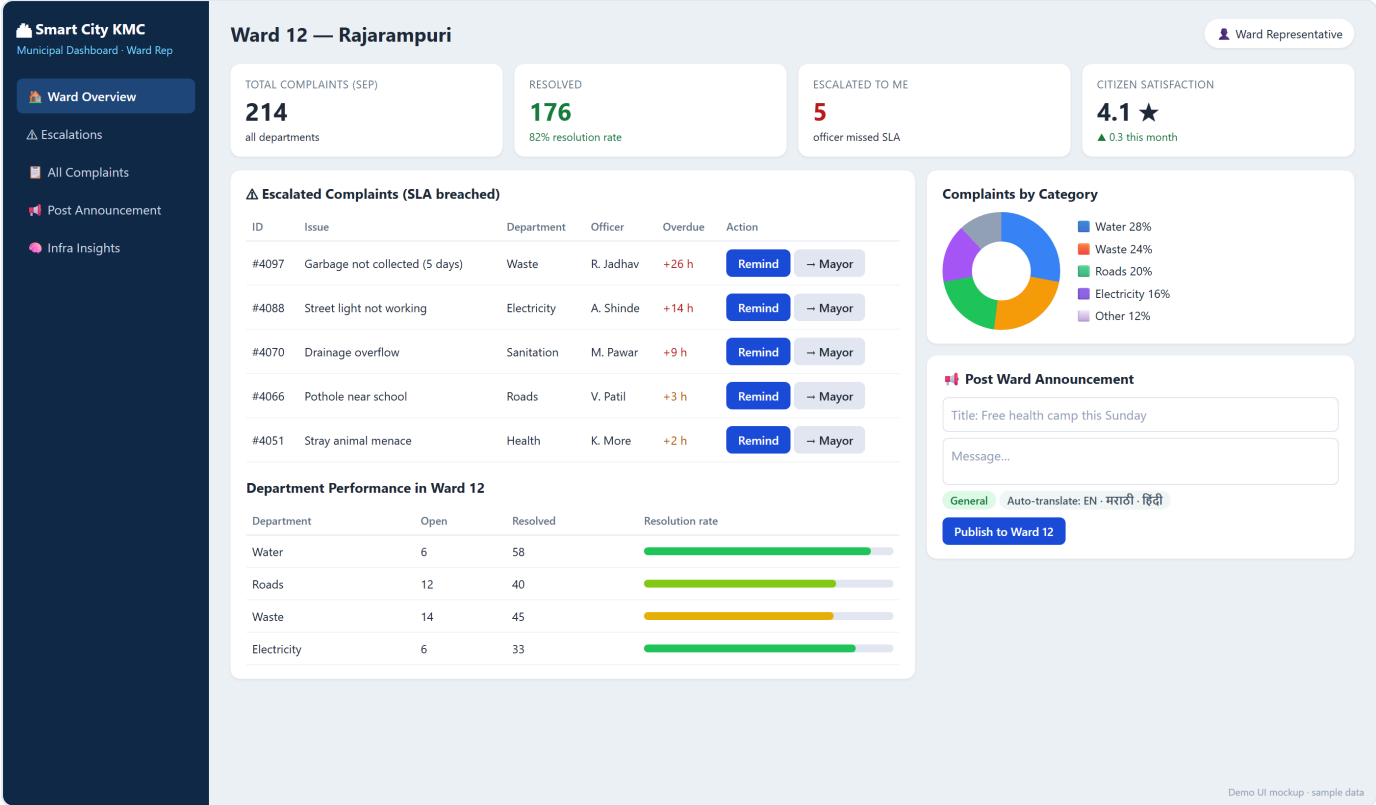


Figure 5: Ward Representative view. Escalated complaints and department performance in the ward.

- Sees **all departments' complaints in their own ward**.
- **Escalation inbox:** complaints where the officer missed the deadline arrive here automatically. From there the rep can send a reminder or escalate to the Mayor.
- Department performance in the ward, complaints by category, and citizen satisfaction.
- Post ward-level announcements with automatic translation.

3.4 Mayor View

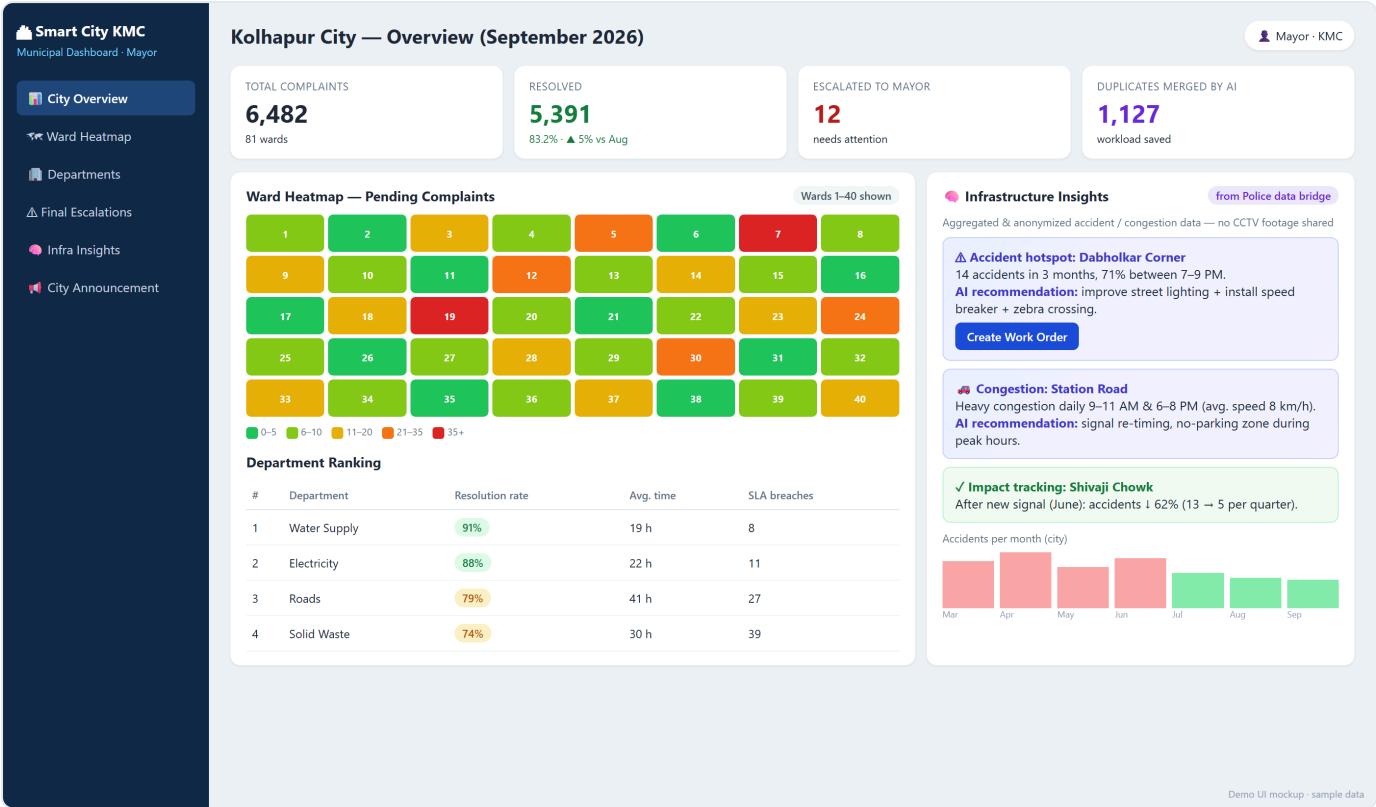


Figure 6: Mayor view. City KPIs, ward heatmap, department ranking and infrastructure insights from the police bridge.

- **City-wide KPIs:** total complaints, resolved, final escalations, duplicates merged by AI.
- **Ward heatmap** of pending complaints, to spot wards that need attention.
- **Department ranking** by resolution rate, average time and deadline misses.
- **Infrastructure Insights** built from police bridge data:
 - Accident hotspots and congestion patterns
 - AI recommendations (e.g., street lighting, speed breaker, signal re-timing)
 - **Impact tracking:** did accidents drop after the fix?
 - One-click work order creation
- City-wide announcements.

3.5 Public Statistics (common to citizens and authorities)

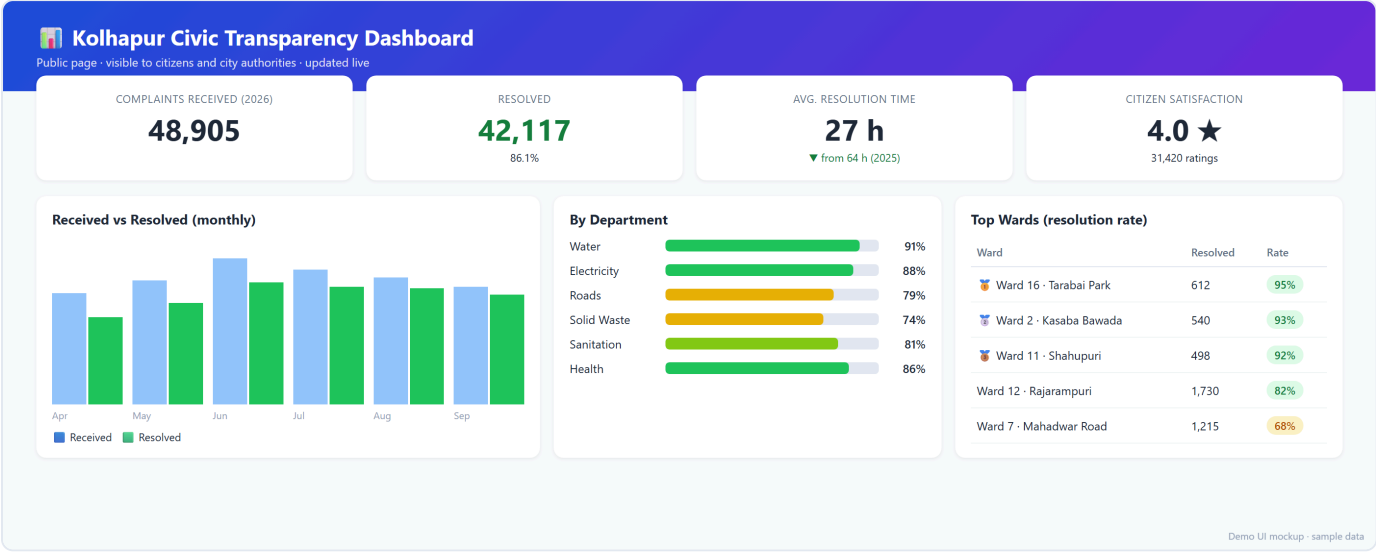


Figure 7: Public statistics page. Complaints received vs resolved, by department and ward.

- Total complaints, resolved count and rate, average resolution time, citizen satisfaction.
- Monthly received vs resolved trend, department-wise resolution rates, ward rankings.
- Holds departments publicly accountable and builds citizen trust.

3.6 Announcement Module

Feature	Description
Ward-specific delivery	Citizens receive only announcements for their ward.
Who can post	Dept. officer (own department), Ward rep (own ward), Mayor (whole city), Police road events (through the bridge).
Priority levels	● Emergency (push + SMS, pinned) · ● Important (push) · ● General (feed only)
AI-drafted announcements	Agents draft notices from events (e.g., pipeline repair). The officer approves them with one click.
Auto-translation	English, Marathi and Hindi, generated by an LLM.
Recurring schedules	E.g., the daily water supply timetable.
Linked to complaints	New complaints explained by an active announcement (e.g., "no water" during a scheduled shutdown) get an automatic reply, which reduces duplicate complaints.

3.7 Accountability and Escalation



- Deadlines depend on category and priority (e.g., pipeline burst 6 h, pothole 48 h, streetlight 72 h).
- A complaint can't be closed without an after-photo that AI has checked. The citizen can reopen it.
- All escalations and deadline misses are counted in department performance and the public statistics.

4. System 2: Police Control Room

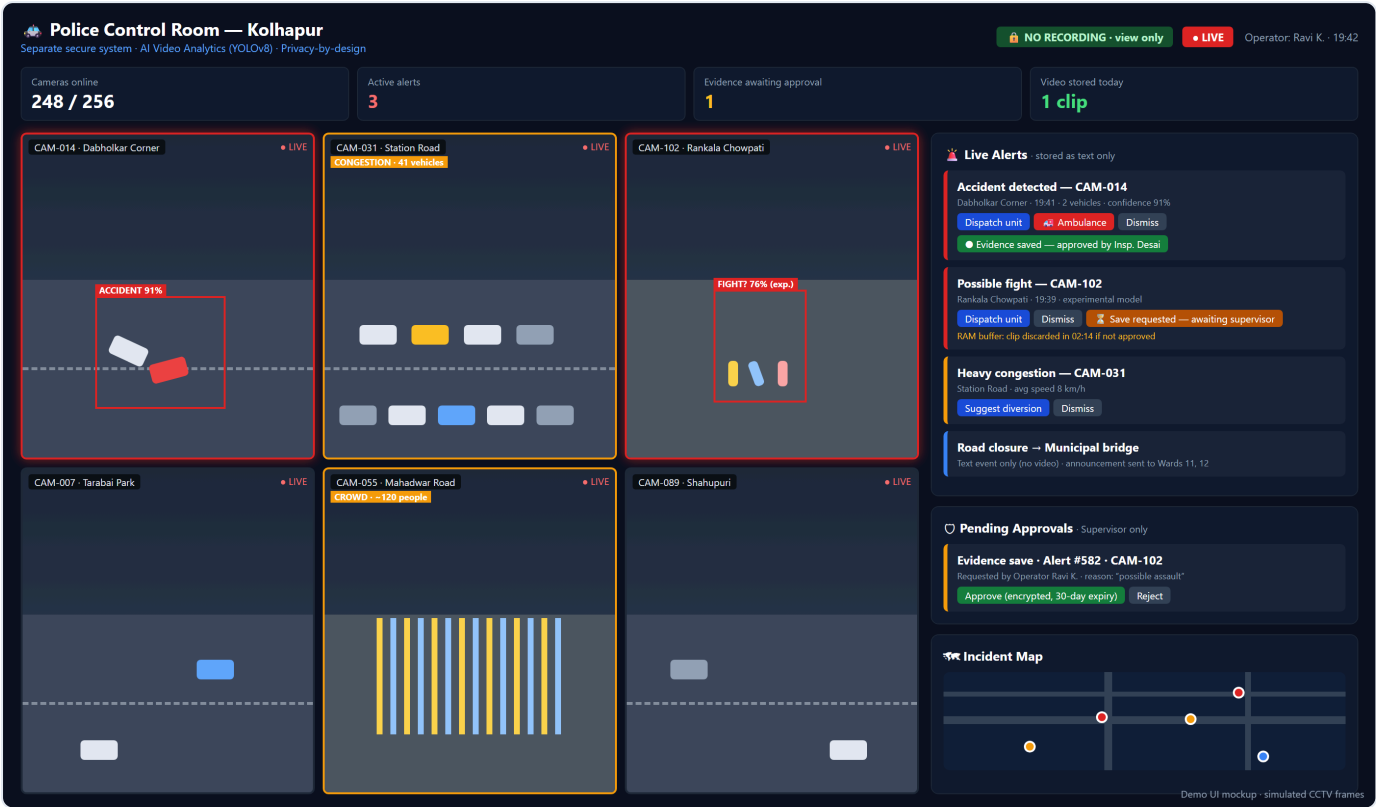


Figure 8: Police Control Room. Live CCTV with AI detections, text-only alerts, evidence approval and incident map (simulated frames).

Feature	Description
CCTV grid	Multiple camera feeds (recorded video or webcam in the prototype). Cameras with alerts are highlighted.
Accident detection	YOLOv8 detects collisions and overturned vehicles, then raises a red alert with confidence and location.
Congestion detection	Vehicle counting and speed estimation per camera. Yellow alert with a suggested diversion.
Crowd detection	Person counting above a threshold gives a crowd alert.
Fight / violence detection	Action recognition on video clips. <i>Experimental</i> ; its accuracy is reported honestly.
Alert actions	Dispatch a unit, notify an ambulance, suggest a diversion, dismiss, or request evidence save .
Pending approvals	Supervisor panel to approve or reject evidence-save requests.
Incident map and history	Map of active incidents plus a searchable incident log.
Security	Separate server, database and login on the police network. No recording by default; footage never leaves the police system.

4.1 Privacy-by-Design: "View live, keep nothing by default"

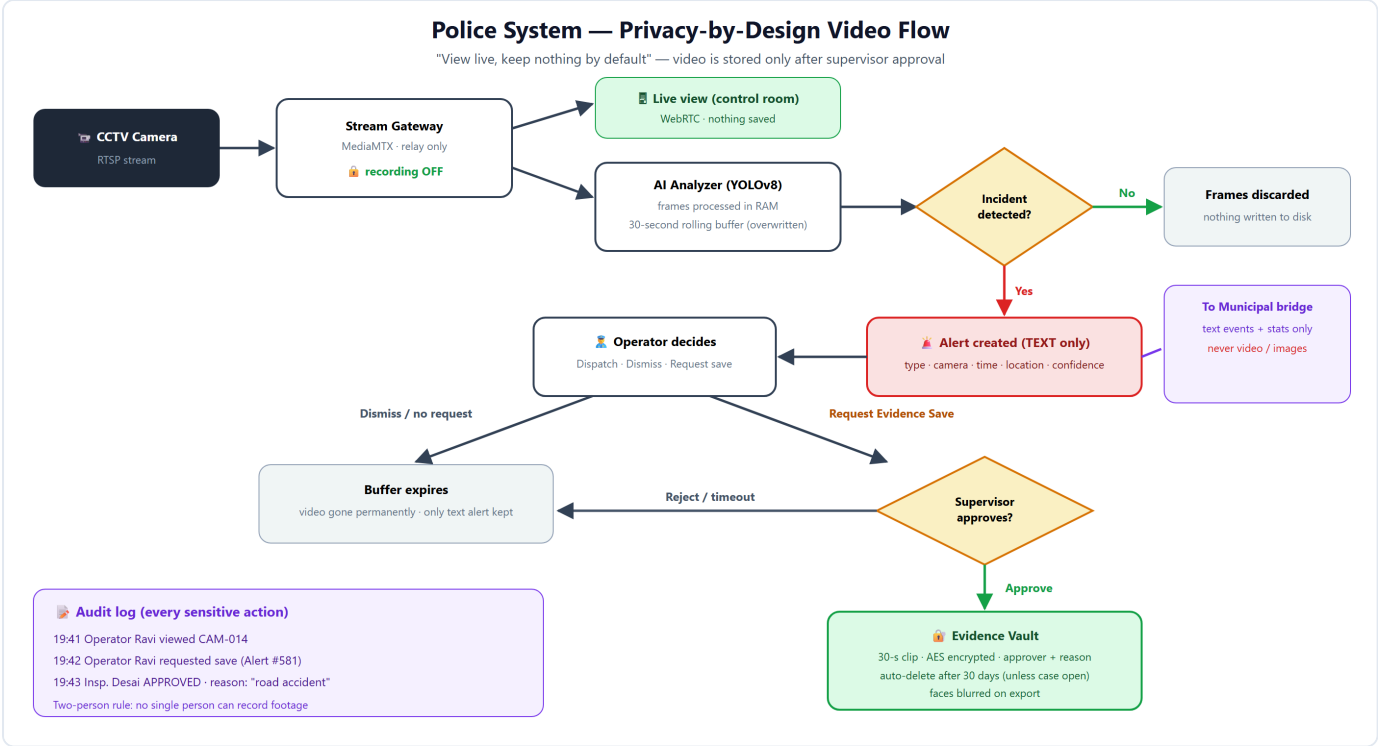


Figure 9: Police video flow. Video is never stored unless a supervisor approves.

Data	Default behaviour	Stored?
Live video	Relayed to the control room screen only (MediaMTX → WebRTC, recording off)	✗ Never
AI processing	Frames analysed in RAM; a 30-second rolling buffer is overwritten continuously	✗ Never
Alert details	Type, camera, time, location, confidence	✓ Text only
Evidence clip	Saved only after supervisor approval; AES-encrypted; auto-deleted after 30 days unless a case is open	⚠ With approval
Data to municipal	Aggregated statistics and text events	✗ No video or images

Who can do what

Role	Live view	Alerts	Request save	Approve save	View saved clips
Control room operator	✓	✓	✓	✗	✗
Supervisor / Inspector	✓	✓	✓	✓	✓
IT Admin	✗	✗	✗	✗	✗ (manages system only)

- **Two-person rule:** an operator can only *request* a save, and a supervisor must approve it. If nobody approves before the buffer runs out, the video is gone permanently.
- **Audit log:** every view, save request, approval and clip access is recorded with user and time.
- **Face blurring** on any exported clip (OpenCV).
- **Network isolation:** the police dashboard is reachable only inside the police network.

- Follows the principles of India's **Digital Personal Data Protection (DPDP) Act, 2023**: purpose limitation, data minimisation and storage limitation.

```
buffer = deque(maxlen=30 * FPS)          # last 30 s, RAM only
while True:
    ok, frame = cap.read()
    buffer.append(frame)                  # never written to disk
    result = model(frame)                 # YOLOv8
    if is_incident(result):
        create_alert(camera_id, result) # stores TEXT only
# a clip is written (encrypted) ONLY when a supervisor approves the save request
```

5. Privacy Data Bridge (Police → Municipal)

Shared (aggregated / event only)	Used by the Municipal Corporation for
Accident hotspots (location + count per month)	Speed breakers, signals, lighting, junction redesign
Congestion patterns (road + time + severity)	Road widening, one-way / no-parking plans, signal timing
Road closures / diversions	Automatic citizen announcements in affected wards
Infrastructure issues seen on CCTV (waterlogging, fallen tree, broken signal)	Automatic complaint created for the right department

Never shared: video footage, faces, number plates, crime or fight details, personal data.

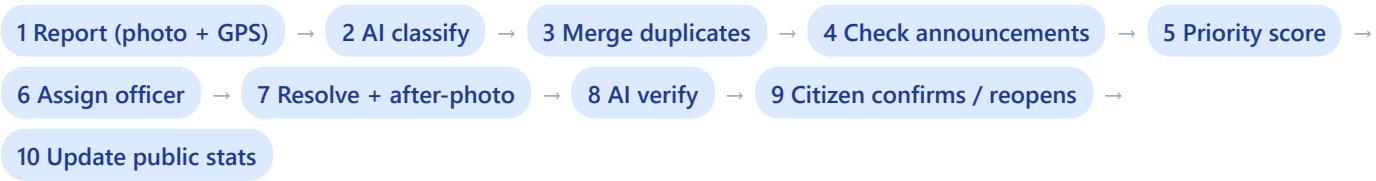
6. AI Agents and Models

Agent	System	Responsibility	Model / Technique
Orchestrator	Municipal	Routing, priority scoring, duplicate merging, coordinating agents	LangGraph + rules/ML scoring
Complaint Classifier	Municipal	Issue type, department, location, severity from text and photo	Sentence-BERT (all-MiniLM-L6-v2), NER, LLM
Road & Waste	Municipal	Detect potholes and garbage; check before/after photos	YOLOv8
Utilities	Municipal	Forecast water and power demand; detect leaks and theft	LSTM, Isolation Forest
Announcement	Municipal	Draft, translate and link announcements to complaints	LLM (Gemini / Llama / GPT)
Escalation	Municipal	Deadline monitoring, officer → ward rep → mayor	Rule-based scheduler
Infrastructure Insights	Municipal	Hotspot analysis, recommendations, impact tracking	Clustering (DBSCAN) + LLM
Video Analytics	Police	Accident, congestion, crowd, fight detection	YOLOv8 + action recognition
Alert & Dispatch	Police	Alert creation, dispatch suggestions, bridge events	Rules + LLM

6.1 Priority Score (example)

Priority = 0.35·Severity + 0.20·Duplicate count + 0.20·Location sensitivity (school/hospital) + 0.15·Time pending + 0.10·Forecast risk. This gives a score from 0 to 100 that sorts the officer's queue.

7. Complaint Workflow



If the deadline is missed at step 6 or 7, the Escalation Agent moves the complaint up the hierarchy.

7.1 Demo Scenario: Road Accident

1. CCTV at Dabholkar Corner detects an accident (Police system) → red alert in the control room.
2. The operator dispatches a unit and an ambulance, then closes the road.
3. The road closure goes through the bridge → the municipal system auto-publishes a **citizen announcement** to the affected wards.
4. Road damage at the site → **automatic complaint** to the Roads Department.

5. The accident adds to the monthly **hotspot statistics** → the Mayor's dashboard recommends a speed breaker and lighting → a work order is created.
6. Months later, **impact tracking** shows whether accidents at that junction went down.

8. Technology Stack

Layer	Technology
Citizen mobile app	React Native (Expo), expo-image-picker, expo-location, expo-notifications, i18next
Web dashboards	React.js, Tailwind CSS
Backend	Python FastAPI, Uvicorn, WebSockets, JWT authentication (two separate servers: municipal and police)
Video streaming	MediaMTX (RTSP → WebRTC relay, recording disabled), OpenCV
Security	HTTPS, HMAC-signed bridge, AES encryption for approved evidence, audit logging
Database / Storage	PostgreSQL; object storage for images
Computer Vision	YOLOv8 (Ultralytics), OpenCV
ML / Forecasting	PyTorch (LSTM), Scikit-learn (Isolation Forest, DBSCAN)
NLP / LLM	Sentence-Transformers, Hugging Face, Gemini / Llama 3 / GPT API
Agent Orchestration	LangGraph
Notifications	Firebase Cloud Messaging, SMS gateway (optional)
Tools	Git & GitHub, Postman, Docker, Google Colab (model training)
Data	Public datasets (RDD2022, garbage and accident datasets), simulated sensor data, recorded CCTV clips

9. Novelty

1. **Agentic AI orchestration.** Autonomous agents detect, prioritize, route and escalate issues across municipal departments without manual coordination.
2. **Privacy-preserving police → municipal bridge.** Aggregated CCTV analytics (never footage) feed infrastructure planning, with AI recommendations and before/after impact tracking.
3. **Closed-loop accountability.** Deadline-based escalation through the ward hierarchy, AI-checked resolution proof, and public statistics.
4. **Two-way citizen communication.** Agent-drafted, multilingual, ward-specific announcements linked to incoming complaints, so fewer duplicate complaints are filed.
5. **AI priority scoring and duplicate merging** instead of first-come-first-served handling.
6. **Privacy-by-design police video.** Live viewing with in-memory AI analysis; footage is stored only after supervisor approval (two-person rule), encrypted, auto-expiring and audit-logged. Most CCTV systems record everything continuously.

9.1 Comparison with Existing Systems

Feature	City complaint apps	Police command centres (ICCC)	Single-domain AI research	Proposed system
Photo complaints + tracking	✓	✗	✗	✓
AI classification + priority scoring	Partial	✗	Partial	✓
Duplicate merging	✗	✗	✗	✓
Auto escalation (officer → ward → mayor)	Partial	✗	✗	✓
AI-checked resolution proof	✗	✗	✗	✓
Announcements linked to complaints	✗	✗	✗	✓
AI CCTV incident detection	✗	Partial	✓	✓
CCTV analytics → infrastructure planning	✗	✗	Partial	✓
No recording by default; evidence saved only with approval	✗	✗	✗	✓
Agentic multi-agent orchestration	✗	✗	✗	✓

Novelty statement: Unlike existing smart city systems that monitor domains in isolation and depend on human coordination, the proposed system uses an agentic AI orchestration layer that autonomously detects, prioritizes and coordinates civic issues. It adds hierarchical accountability with AI-checked resolution, two-way multilingual communication with citizens, a privacy-preserving bridge that turns police CCTV analytics into data-driven infrastructure planning, and a privacy-by-design police video pipeline that stores footage only with supervisor approval.

Estimated novelty (the team's subjective estimate): about 45% for the whole project and about 65–70% for the core contribution (orchestration, bridge, accountability and the announcement loop). Most standard models (YOLO, LSTM, Isolation Forest) are existing techniques.

10. Project Schedule (16 Weeks)

Weeks	Work
1–2	Literature review, study of existing systems (ICCC, complaint apps), requirements, novelty comparison
3–4	Architecture, database design, role-based authentication, project setup
5–6	Citizen mobile app (React Native): photo complaint, GPS, tracking, push notifications; NLP complaint classification
7–8	YOLOv8 models: potholes and garbage (municipal); accidents, congestion and crowds (police)
9–10	Utilities agent: simulated sensors, LSTM forecasting, Isolation Forest anomalies
11–12	LangGraph orchestrator, priority scoring, duplicate merging, escalation, announcements
13–14	Municipal dashboards (officer / ward / mayor), police control room with live view and evidence approval, privacy bridge, public statistics
15–16	Testing, measured comparison with manual routing, report, presentation, deployment

11. Suggested Work Distribution

Member	Module
Paras Pilankar	Orchestrator agent (LangGraph), backend APIs, escalation logic
Savej Shaikh	Computer vision: YOLOv8 for complaints and police CCTV analytics; live streaming and in-memory buffer
Harsh Vanjare	Utilities agent (LSTM, Isolation Forest), sensor simulation, privacy bridge and insights
Shanur Shaikh	Municipal dashboards (officer, ward rep, mayor) and police control room UI
Pranav Sangar	Citizen mobile app (React Native), announcement module, NLP classification, public statistics page

Suggested split only. Adjust it to each member's strengths.

12. Expected Outcomes

- A working prototype of 3 dashboards (Municipal with 3 role views, Police, Citizen) plus a public statistics page.
- Faster complaint routing and response than manual routing, **measured** in simulated scenarios.
- Fewer duplicate complaints through AI merging and announcement linking.
- Transparent, accountable governance through escalation, verified resolution and public statistics.
- Data-driven infrastructure decisions from privacy-preserving police analytics.

Evaluation Metrics

- Detection accuracy: mAP, precision and recall for the YOLO models; F1 score for complaint classification
- Forecast error: MAE/RMSE for the LSTM; precision for anomaly detection

- Routing time and escalation effectiveness vs a manual baseline
- Number of duplicate complaints prevented; dashboard usability (user survey)

13. Limitations and Assumptions

- CCTV and IoT data are recorded or simulated. There is no live city feed in the prototype.
- Fight/violence detection is experimental and may have lower accuracy.
- Police and ambulance dispatch is simulated through alerts.
- The evidence approval flow is demonstrated with sample users; real deployment would follow police legal procedures for evidence handling.
- Production deployment would require government network security, data-protection compliance (DPDP Act) and integration with existing municipal systems.

All screens in this document are demo UI mockups with sample data, made for illustration.